

Curriculum Vitae

PERSONAL INFORMATION

Walter Lucas, Ph.D.
University of Brasília
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Birth date: 15 May 1992
Place of birth: Brasília, Brazil
Nationality: Brazilian

EDUCATION

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|-------------|---|
| 2019 - 2026 | Ph.D., Computer Science
University of Brasília, UnB, Brazil
Thesis: A Socio-Technical Grounded Theory about Source Code Rejuvenation
Advisor: Rodrigo Bonifácio de Almeida
Co-Advisor: João Alexandre Baptista Vieira Saraiva |
| 2017 - 2019 | Master.Sc., Computer Science
University of Brasília, UnB, Brazil
Thesis: Analysis of the impact on the understanding of Java programs with the adoption of lambda expressions
Advisor: Rodrigo Bonifácio de Almeida |
| 2011 - 2015 | Bachelor's Degree, Information System
University Center of Patos de Minas, UNIPAM, Brazil
Thesis: UNIEXPO: platform for dissemination of extension projects
Advisor: Jean Lucas de Souza |

PROFESSIONAL AND ACADEMIC EXPERIENCES

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| 2024 - Current | Fóton Informática S.A., Brasília, Brazil Position: Java Developer Analyst
Project: Conduction and refinement of requirements, identifying unmapped needs during the development cycle, and proposing solutions to increase efficiency, usability, and functional adherence. Conception and development of new modules and evolutionary improvements for corporate system modernization. Diagnosis and resolution of staging and production incidents, refactoring and optimizing code with SonarLint and Fortify, and implementing automated testing with JUnit and Mockito.
Technologies: Java 11/17, JUnit, SCRUM, Microservices, Spring Framework, Maven, Java EE, Java SE, Tomcat, Jira, DBeaver, GIT, Oracle, Design Patterns, REST APIs, Swagger, Mockito, SonarLint, Fortify. |
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- 2023 - 2024 **University of Brasília, UnB, Brasília, Brazil** Position: Developer and Product Owner (Virtual Social Office - Phase 2)
 Project: Developed new features to facilitate information sharing among partner institutions, coordinating backend, web frontend, and mobile teams. Led team selection and management, software architecture design and validation, and Expo/React Native app migration. Integrated chatbots, authorization systems, and conducted Google Analytics UX studies.
 Technologies: React Native, Expo, Node.js, Java 21, Spring Boot 3, Google Analytics, Microservices, LGPD Compliance, ETL, Pentaho, REST API, Docker, GitHub, Scrum.
- 2022 - 2023 **INESC TEC International Visiting Researcher Programme, Portugal**
 Position: Researcher
 Project: Conducted an in-depth study analyzing GitHub repository histories of 272 projects to understand KDE contributor practices when adopting modern C++ features.
 Research topic: Software engineering, C++ evolution, Software rejuvenation, Mining Software Repositories
 Supervisor: João Saraiva - University of Minho - Portugal
- 2020 - 2022 **University of Brasília / Finatec, Brasília, Brazil** Position: Data Engineer (Virtual Social Office - Phase 1)
 Project: Developed and implemented ETL solutions to store and validate data collected by Social Offices, ensuring information quality for the ESVirtual application. Conducted pilots across four Federation Units (BA, ES, PB, RJ) and automated spreadsheet workflows.
 Technologies: ETL, Python, Data Management, Information Security, User-Centered Design, Java, Mobile Applications, Sprint Planning
 Supervisor: Rodrigo Bonifácio de Almeida
- 2018 - 2020 **Department of Electrical Engineering, University of Brasília, Brazil**
 Position: Data Engineer / Software Developer
 Project: Applied Research in Strategic Planning, Corporate Risk Management and Corporate Information Management for the Ministry of Planning (DIPLA/MP). Maintained and updated management dashboards using ETL tools like Pentaho and QlikView, created OLAP cubes, implemented Qlik Sense dashboards with HTML/CSS/jQuery, and managed Rancher containers and Jenkins CI/CD pipelines.
 Technologies: Java, PHP, Pentaho, QlikView, Qlik Sense, OLAP, HTML, CSS, jQuery, JavaScript, Rancher, Jenkins, CI/CD, ETL, Risk Management, Cloud Computing
 Supervisor: Rafael Tímoteo de Sousa Junior
- 2017 - 2018 **Foundation for Scientific and Technological Enterprise - Finatec, Brazil**
 Position: Software Developer (PromiseEB Project)
 Project: Investigated software modernization practices and methods for the Brazilian Army, utilizing "Software Cities" (CodeCity) and "Overview Pyramids" (iPlasma) visualizations. Applied BDD testing and agile methodologies for system reengineering.
 Technologies: Java, CodeCity, iPlasma, BDD, Reverse Engineering, Reengineering, Scrum, Agile Methodologies, Software Visualization
 Supervisor: Rodrigo Bonifácio de Almeida
- 2016 - 2017 **Mídia Proativa, Patos de Minas, MG, Brazil**
 Position: PHP Developer
 Project: Developed and managed e-commerce and web systems, serving as manager and developer for online and physical point-of-sale systems during the Na-

tional Corn Festival (Fenamilho 2016). Performed requirements elicitation and maintained PHP systems using CakePHP 2.x and Laravel 5/6.
Technologies: PHP, CakePHP 2.x, Laravel 5/6, MySQL, HTML, CSS, jQuery, JavaScript, Mobile Applications, Project Management, Requirements Elicitation

2015 - 2015

MAXPHONE, Patos de Minas, MG, Brazil

Position: Systems Analyst

Project: Developed web solutions using PHP/CakePHP and MySQL. Analyzed data to optimize product sales via Google Analytics, performed continuous website maintenance, and managed digital marketing and promotional banners.

Technologies: PHP, CakePHP 2.x, MySQL, Google Analytics, HTML, CSS, jQuery, JavaScript, Facebook Page Management

PUBLICATIONS

Journal Publications

1. MENDONÇA, W. L. M., FORTES, J., LOPES, F. V., MARCÍLIO, D., DE ALMEIDA, R. B., CANEDO, E. D., LIMA, F., AND SARAIVA, J. Understanding the impact of introducing lambda expressions in java programs. *Journal of Software Engineering Research and Development* 8 (2020), 7–1
2. LUCAS, W., CARVALHO, F., NUNES, R. C., BONIFÁCIO, R., SARAIVA, J., AND ACCIOLY, P. Embracing modern c++ features: An empirical assessment on the kde community. *Journal of Software: Evolution and Process* 36, 5 (2024), e2605
3. LUCAS, W., NUNES, R. C., BONIFÁCIO, R., CARVALHO, F., LIMA, R., SILVA, M., TORRES, A., ACCIOLY, P. R. G., MONTEIRO, E., AND SARAIVA, J. Understanding the adoption of modern javascript features: An empirical study on open-source systems. *Empir. Softw. Eng.* 30, 3 (2025), 107

Conferences

1. LUCAS, W., BONIFÁCIO, R., CANEDO, E. D., MARCÍLIO, D., AND LIMA, F. Does the introduction of lambda expressions improve the comprehension of java programs? In *Proceedings of the XXXIII Brazilian Symposium on Software Engineering* (2019), pp. 187–196
2. DE MENDONÇA, W. L. M., COSTA, P. H. T., CANÇADO, E. C. R., LIMA, F., CANEDO, E. D., BONIFÁCIO, R., AND AMARAL, L. H. V. From dusk till dawn: Reflections on the impact of covid-19 on the development practices of a r&d project. In *Proceedings of the 34th Brazilian Symposium on Software Engineering* (2020), pp. 596–605
3. DA FROTA, F. V. L., DE MENDONÇA, W. L. M., DA SILVA, D. A., DE SOUZA FERNANDES, M. A., BARBOSA, N. F., AND CANEDO, E. D. Utilização de algoritmos genéticos em gerações de dados de teste de software com agrupamento de genes e redução de espaço de busca. *WWW/INTERNET 2019*, 87

Extended Abstract / Short Papers

1. DANTAS, R., CARVALHO, A., MARCILIO, D., FANTIN, L., SILVA, U., LUCAS, W., AND BONIFÁCIO, R. Reconciling the past and the present: An empirical study on the application of source code transformations to automatically rejuvenate java programs. In *2018 IEEE 25th international conference on software analysis, evolution and reengineering (SANER)* (2018), IEEE, pp. 497–501
2. LUCAS, W., BONIFÁCIO, R., AND SARAIVA, J. Understanding the motivations, challenges, and practices of software rejuvenation. In *2023 IEEE International Conference on Software Maintenance and Evolution (ICSME)* (2023), pp. 611–616

HONORS, AWARDS & SCHOLARSHIPS

2019	Best paper award in Proceedings of the XXXIII Brazilian Symposium on Software Engineering (SBES).
2016	Unideias, Patos de Minas University Center.
2015	7th Entrepreneurship Award, Patos de Minas University Center.

LICENSES & CERTIFICATIONS

2026	AWS Certified AI Practitioner - Amazon Web Services (AWS)
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LANGUAGES

Portuguese (Native)
English (Advanced, TOEFL ITP: 443)
Spanish (Intermediate)